

Construction and Unit Test Report

ExpenseMate – Personal Expense Manager

Software Construction, Assignment 1 | Phase 3: Coding & Unit Testing | Version 1.0, 27 September 2026

1. Modules

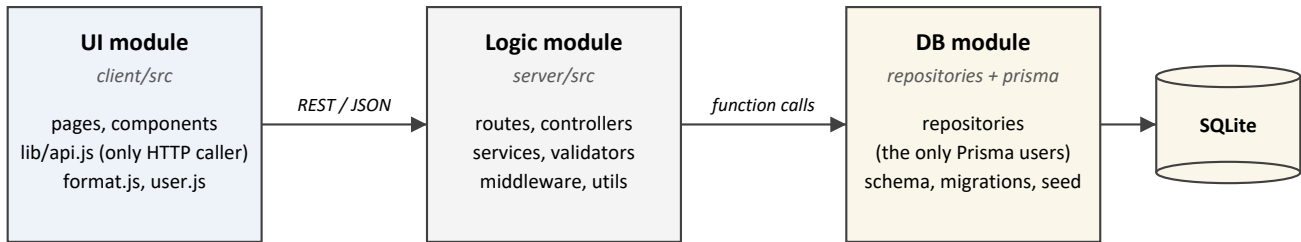


Figure 1. Modules and their interfaces

Module	Location	Responsibility	Size
UI	client/src	Screens, forms, charts; calls the API only through lib/api.js	35 files, 5,046 lines
Logic	server/src	Login, validation, budgets, analytics, CSV rules, errors	29 files, 699 lines
DB	repositories, prisma	All database reads and writes; schema and migrations	10 files, 409 lines

Lines exclude blank lines and tests; the UI figure includes 2,240 lines of CSS.

2. API Contract (UI ↔ Logic)

Method	Path	Purpose
POST, GET	/api/auth/signup, /login, /logout, /me	Accounts and session
GET, POST, DELETE	/api/categories[:id]	Categories
GET, POST, PATCH, DELETE	/api/transactions[:id]	Transactions (filter by month and type)
GET, POST	/api/transactions/export, /import	CSV export and import
GET, PUT, DELETE	/api/budgets[:id], /api/budgets/alerts	Budgets and alerts
GET	/api/analytics/summary, /by-category, /trend	Analytics

3. Source Code

Folder	Contents
client/src	pages, components, context, lib (api, format, queries)
server/src	routes, controllers, services, validators, middleware, repositories, utils
server/prisma	schema, migrations, seed data
server/tests, client/src/**/*.test.js	unit and integration tests

Run: npm install and npm run dev in *server* and *client*; npm run db:reset loads demo data; npm test runs the tests.

Practices: layer rules kept; Zod at every entry point; one error handler; money in integer paisa; test-first for the edit endpoint; small commits.

4. Unit Testing

- Logic** (Jest 30.5): services tested with mocked repositories; validators and security tested directly.
- UI** (Vitest 5.0): formatting, password strength and API client tested with a stubbed network and a fixed clock.
- DB:** repositories are thin Prisma calls, so they are covered by integration tests (Phase 4).
- Boundary cases tested: exactly 80% and 100% of a budget, December → January, 31 February, 1,000 vs 1,001 CSV rows.

Module	Test file	Unit under test	Tests	Passed	Failed
UI	format.test.js	lib/format.js	15	15	0
	user.test.js	lib/user.js	7	7	0
	api.test.js	lib/api.js	9	9	0

Module	Test file	Unit under test	Tests	Passed	Failed
Logic	authService.test.js	services/authService.js	9	9	0
	categoryService.test.js	services/categoryServices.js	5	5	0
	transactionService.test.js	services/transactionService.js	8	8	0
	budgetService.test.js	services/budgetService.js	10	10	0
	analyticsService.test.js	services/analyticsService.js	7	7	0
	csvService.test.js	services/csvService.js	6	6	0
	validators.test.js	validators/*.js	11	11	0
	security.test.js	utils/security.js	4	4	0
Total			91	91	0

5. Coverage (unit tests only)

File	Statements	Branches	Functions	Lines
services/analyticsService.js	100%	100%	100%	100%
services/authService.js	95.5%	88.9%	85.7%	95.5%
services/budgetService.js	100%	91.7%	100%	100%
services/categoryServices.js	93.3%	100%	66.7%	93.3%
services/csvService.js	100%	100%	100%	100%
services/transactionService.js	92.3%	75%	100%	92.3%
validators (all)	100%	100%	100%	100%
utils/security.js	100%	100%	100%	100%
Logic total	95.6%	89.9%	94.4%	95.5%
UI: lib/format.js	88.6%	100%	77.8%	88.6%
UI: lib/user.js	96.4%	90%	100%	96.4%
UI: lib/api.js	67.2%	80.5%	41.4%	70%

React components and data hooks are not unit tested; they are covered by system tests in Phase 4.

6. Defects Found During Construction

#	Defect	Found by
1	End date built with <i>DataTransfer</i> instead of <i>Date</i> ; monthly queries crashed	Unit test
2	Import of a non-existent file; server would not start	Code review
3	Export name mismatch; listing crashed	Code review
4	Routes mounted after the error handler; HTML 500 instead of JSON	Code review
5	Wrong enum value "Expense"; budgets API returned 500	Integration test
6	Transaction DELETE route missing <i>:id</i>	Integration test
7	Transaction count queried the wrong table; category delete crashed	Integration test
8	Category DELETE route missing <i>:id</i>	Integration test

All eight were fixed and each is now covered by a test.